

Final Report - Project 7

Sentiment Analysis

Group 19

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Abstract

This project studies binary sentiment classification on the Amazon Reviews dataset. Reviews are classified as negative or positive using TF-IDF with Logistic Regression, a configurable LSTM/BiLSTM neural network, and reduced RoBERTa fine-tuning experiments. The final experiments focus on reproducibility, dataset-size reduction, hyperparameter sensitivity, regularization, and computational cost. Logistic Regression remains the strongest practical baseline, reaching 94.07% accuracy on the full test set. On a 100k-sample experiment, Logistic Regression reaches 91.34% test accuracy in 11.6 seconds, while the best LSTM configuration reaches 90.71% but requires around 9 minutes. The best reduced RoBERTa experiment reaches 94.25% test accuracy on a 5k-sample setup, but requires substantially more runtime than Logistic Regression.

1 Introduction

Sentiment analysis is a standard Natural Language Processing task where the goal is to automatically detect the polarity of a text. In this project, we work on binary sentiment classification of Amazon product reviews. The label 0 represents negative reviews and the label 1 represents positive reviews.

The project compares methods of increasing complexity. Logistic Regression with TF-IDF is used as a fast and interpretable baseline. LSTM models are used to test whether sequence modeling improves performance. A RoBERTa pipeline is used to evaluate whether partial transformer fine-tuning improves accuracy enough to justify its additional cost.

2 Dataset and Preprocessing

We use the Amazon Reviews dataset from Kaggle by Adam Bittlingmayer. The dataset is already formatted for fastText:

- `__label__1`: negative review, mapped to class 0;

- `__label__2`: positive review, mapped to class 1.

The full dataset contains 3.6 million training reviews and 400,000 official test reviews. The classes are balanced, so accuracy is a meaningful metric.

The training file is split into train and validation subsets using a stratified 80/20 split. This preserves the 50/50 class balance. All experiments use seed 42 unless otherwise stated. The code fixes Python, NumPy, and PyTorch random seeds, and passes the seed to the model-specific training components.

The shared preprocessing removes HTML tags, lowercases the text, and normalizes whitespace. Punctuation is kept because it may carry sentiment information.

3 Implemented Methods

3.1 Logistic Regression with TF-IDF

The baseline uses a `scikit-learn` pipeline composed of:

- TF-IDF vectorization with unigrams and bigrams;
- a maximum vocabulary size of 100,000 features;
- `min_df=2` to remove very rare terms;
- sublinear term-frequency scaling;
- Logistic Regression with ℓ_2 regularization and $C = 1.0$.

This approach does not model long-distance word order, but it is very efficient on large text datasets and gives a strong reference point.

3.2 LSTM / BiLSTM

The LSTM implementation contains:

- a vocabulary built from the training set;
- integer encoding and padding/truncation to a fixed sequence length;
- an embedding layer;
- one or more LSTM layers, optionally bidirectional;
- global max pooling over the sequence dimension;
- dropout;
- a final linear classifier.

The model is trained with Adam and cross-entropy loss. We added several hyperparameters to support final experiments: learning rate, embedding size, hidden size, number of layers, bidirectionality, dropout, weight decay, frozen embedding layer, and seed-controlled data loading. The training script writes a `metrics.json` file and, for neural models, a `training_history.json` file.

3.3 RoBERTa

The RoBERTa pipeline uses the Hugging Face Transformers library with `roberta-base`. It supports full fine-tuning, freezing the first N encoder layers, or freezing the whole encoder and training only the classification head. Because full fine-tuning is expensive, the final RoBERTa experiments freeze a large part of the encoder and fine-tune only the upper transformer layers plus the classification head. This makes RoBERTa feasible on the available laptop while still allowing the model to adapt to the sentiment task.

4 Experimental Protocol

We evaluate models with accuracy, macro F1-score, macro precision, and macro recall. Runtime is also recorded because the professor specifically asked us to compare whether additional complexity improves accuracy enough to justify the computation time.

The experiments are organized around the following questions:

- How much can the dataset be reduced while still obtaining useful performance?
- Does LSTM improve over Logistic Regression?
- How sensitive is LSTM to the learning rate?
- Does increasing model capacity improve results?
- Do dropout and weight decay reduce overfitting or improve accuracy?

5 Results

5.1 Dataset Reduction with Logistic Regression

Table 1 shows that Logistic Regression performs well even with a small fraction of the training set.

Table 1: Logistic Regression results when reducing the training set size.

Loaded train examples	Test examples	Test accuracy	Runtime (s)
2,000	2,000	0.8145	4.25
10,000	5,000	0.8698	2.27
100,000	20,000	0.9134	11.57
Full dataset	400,000	0.9407	465.00

The first important finding is that only 2,000 loaded training examples already give 81.45% test accuracy. This is above the 70–80% threshold discussed at the midterm. Increasing the dataset to 100,000 examples improves accuracy to 91.34%, and the full-dataset run reached 94.07% in 465 seconds.

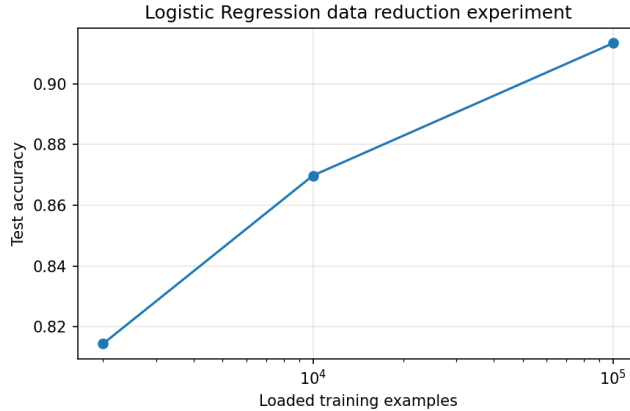


Figure 1: Effect of training-set size on Logistic Regression test accuracy.

5.2 LSTM Hyperparameter Experiments

All LSTM experiments below use 100,000 loaded training examples, 20,000 test examples, 3 epochs, seed 42, and sequence length 256.

Table 2: LSTM experiment results.

Configuration	Test accuracy	Macro F1	Val accuracy	Runtime (s)
BiLSTM, lr= 10^{-3}	0.9071	0.9069	0.9104	535.09
BiLSTM, lr= 10^{-4}	0.8018	0.8001	0.8007	529.24
BiLSTM, 3 layers	0.9070	0.9068	0.9093	1599.01
BiLSTM, dropout=0.5	0.8963	0.8959	0.8967	528.18
BiLSTM, weight decay= 10^{-4}	0.9014	0.9010	0.9023	535.75

5.3 RoBERTa Experiments

Table 3 shows the reduced RoBERTa experiments. The first run freezes the whole encoder and trains only the classification head. The later runs unfreeze the top transformer layers, which gives the model enough capacity to adapt to the sentiment task.

The head-only run is close to random guessing. This is expected because the classification head starts from random weights while the frozen encoder cannot adapt to the task. Once the upper transformer layers are unfrozen, RoBERTa improves sharply. With 5,000 loaded training examples, freezing the first 8 layers and training for 2 epochs reaches 94.25% test accuracy, the best result among our reduced experiments.

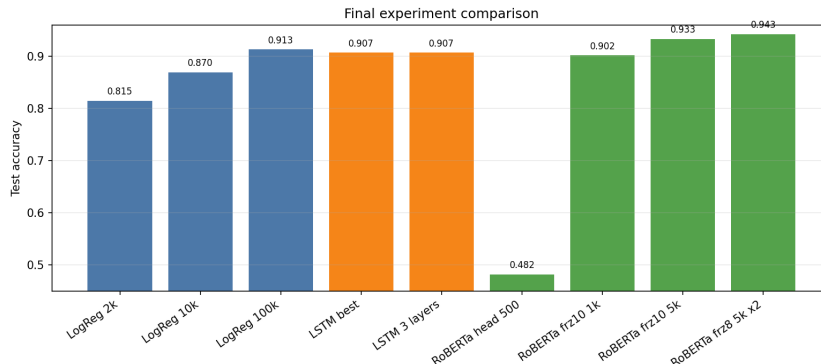


Figure 2: Comparison of selected Logistic Regression, LSTM, and RoBERTa experiments.

Table 3: Reduced RoBERTa fine-tuning results.

Configuration	Train	Test	Epochs	Test accuracy	Runtime (s)
Head only	500	500	1	0.4820	35.35
Freeze 10 layers	1,000	1,000	1	0.9020	70.22
Freeze 10 layers	5,000	2,000	1	0.9330	270.53
Freeze 8 layers	5,000	2,000	1	0.9380	352.34
Freeze 8 layers	5,000	2,000	2	0.9425	707.03

6 Discussion

6.1 Learning Rate

The learning rate has a strong effect on LSTM performance. With $lr = 10^{-3}$, the model reaches 90.71% test accuracy. With $lr = 10^{-4}$, the test accuracy drops to 80.18%. The validation history shows that the smaller learning rate is still improving after 3 epochs, so the problem is likely undertraining rather than overfitting. A lower learning rate may require more epochs to converge.

6.2 Model Capacity

Increasing the LSTM from 1 layer to 3 layers does not improve test accuracy: 90.70% versus 90.71%. Runtime increases from about 535 seconds to 1599 seconds. This is a clear negative result: extra depth increases compute time without meaningful accuracy gain. It may also increase optimization difficulty and risk of overfitting if trained longer.

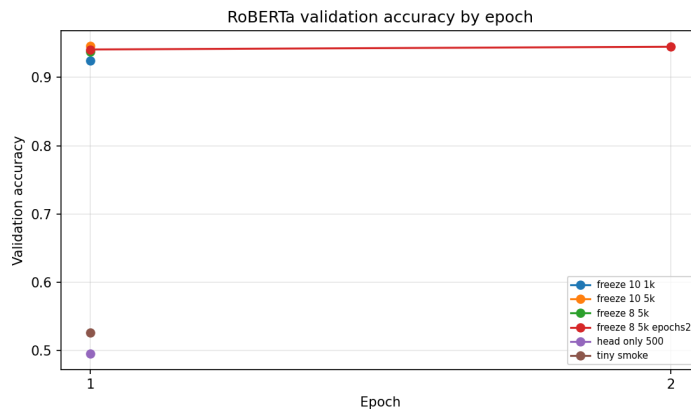


Figure 3: Validation accuracy curves for the RoBERTa experiments.

6.3 Regularization

Increasing dropout from 0.3 to 0.5 reduces test accuracy from 90.71% to 89.63%. Adding weight decay of 10^{-4} reaches 90.14%, also below the baseline LSTM. In these runs, stronger regularization did not improve the final score. The baseline LSTM does not show strong overfitting after 3 epochs because validation accuracy continues to improve while training loss decreases.

6.4 Logistic Regression vs LSTM

The most important conclusion is that Logistic Regression is still the best accuracy/time trade-off. On the 100k experiment, Logistic Regression reaches 91.34% test accuracy in 11.57 seconds. The best LSTM reaches 90.71% but takes around 535 seconds. Therefore, on this dataset, the additional sequence-modeling capacity of LSTM does not justify its computational cost.

This result is explainable. Amazon sentiment reviews often contain strong local lexical cues such as “great”, “bad”, “waste of money”, or “not good”. TF-IDF with bigrams captures many of these patterns efficiently. LSTM can model word order, but the gain is limited because the dataset is large and the baseline features are already very informative.

6.5 RoBERTa Trade-off

RoBERTa becomes the strongest reduced-data model once upper layers are unfrozen. The best RoBERTa run reaches 94.25% test accuracy with only 5,000 loaded training examples, which is higher than Logistic Regression on 100,000 loaded examples. However, this comes at a much higher computational cost: the 2-epoch RoBERTa run takes 707 seconds, compared with 11.57 seconds for Logistic Regression on 100,000 examples. This shows that transformer fine-

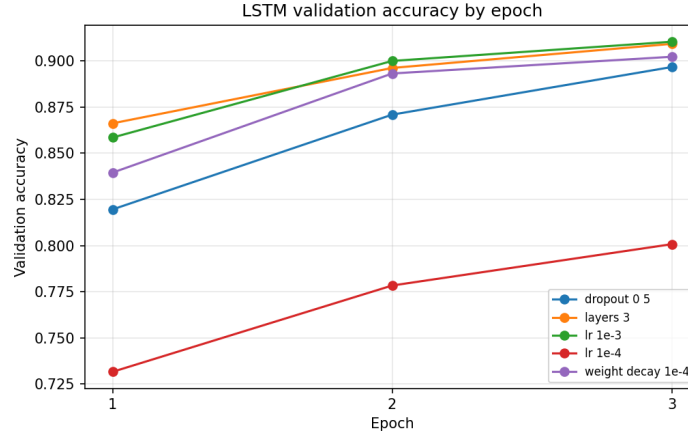


Figure 4: Validation accuracy curves for the LSTM experiments.

tuning can improve accuracy, but the improvement must be weighed against runtime and hardware requirements.

7 Limitations

RoBERTa was evaluated only on reduced subsets, not on the full dataset. This means the reported RoBERTa result should be interpreted as a strong reduced-data experiment, not as a full benchmark of the transformer model. Full fine-tuning on the complete dataset would likely require more memory and more time than was available on the laptop.

Another limitation is that we did not perform a full grid search. Instead, we selected targeted experiments based on the midterm feedback: dataset reduction, learning rate, model depth, dropout, and weight decay.

8 Conclusion

The final project implemented a complete reproducible sentiment classification pipeline and compared classical and neural approaches. Logistic Regression with TF-IDF is a very strong baseline: it reaches 94.07% accuracy on the full test set in 465 seconds, and already reaches 91.34% with only 100,000 loaded training examples.

The LSTM experiments show that neural sequence models are sensitive to hyperparameters. A learning rate of 10^{-3} works much better than 10^{-4} for 3 epochs. Adding layers, dropout, or weight decay did not improve the result in our experiments. The best LSTM reaches about 90.71% test accuracy, slightly below Logistic Regression on the same 100k setup while being much slower.

RoBERTa gives the best reduced-data accuracy when partially unfrozen, reaching 94.25% with 5,000 loaded training examples. The main conclusion is therefore more nuanced: Logistic Regression gives the best performance-cost trade-off, while RoBERTa gives the best accuracy in our reduced experiments but requires significantly more compute. More complex models must be justified not only by accuracy, but also by runtime, hardware requirements, and practical usefulness.